Inverted MAML Card 1: Round 6 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1792, Mom=0.8, Clamp=±90 LSB

1. Round 6 Summary & Cumulative Comparison (R1–R6)

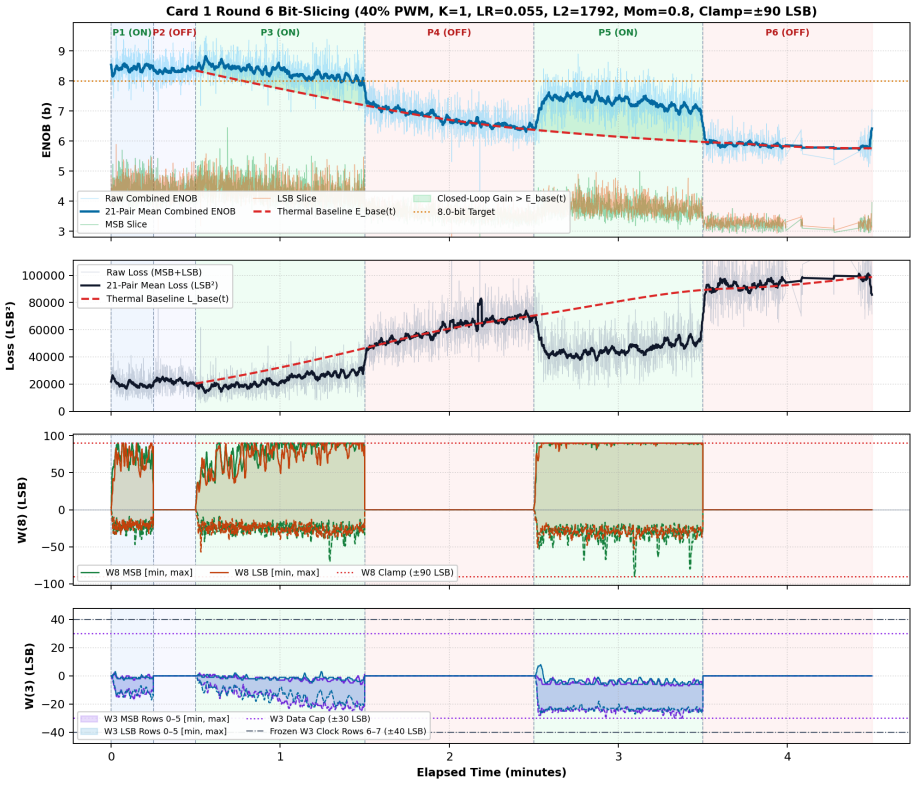
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 6 Detrended Trend Metrics:** Lift vs E_base = **+0.99 b**, Loss Ratio = **1.72x**, Climb = **+1.02 b**, Jitter = **0.40 b**, Recovery = **0.00**, Score = **+1.770**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799
R2dc_fix	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204
R4dc_fix*	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393
R6dc_fix	0.055	1792	0.8	±90	+0.99b	1.72x	+1.02b	0.40b	0.00	+1.770

2. Phase-by-Phase Telemetry (3,828 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	339	8.23->8.46 (8.42b)	-	25098->18668	[90,90]	17	339
P2	Cold / Regler OFF	287	8.36->8.41 (8.36b)	-	22141->20560	[0,0]	0	0
P3	Warm / Regler ON	1044	8.43->8.12 (8.31b)	+0.72 b	18889->27216	[90,90]	25	1044
P4	Warm / Regler OFF	899	7.26->6.51 (6.77b)	+0.03 b	46106->68355	[0,0]	0	0
P5	Warm / Regler ON	819	6.57->7.16 (7.30b)	+1.13 b	66596->49334	[90,90]	30	819
P6	Warm / Regler OFF	440	5.94->5.82 (5.88b)	+0.02 b	91932->96596	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 6

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±90 LSB, while W(3) rows 0–5 stay within ±30 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 6 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

